2023 End Term

Note: Q1 is compulsory. Attempt one question each from Unit I, II, III & IV.

Q1 Conceptually explain the following technical concepts with suitable examples. (2.5*8)

a. Business Process-as-a-Service (BPaaS)
b. Fog Computing
c. Virtual Machine Monitor (VMM)
d. Network Virtualization
e. Threats and Liabilities in Cloud Computing
f. IDaaS (Identity-as-a-Service)
g. Cloud Analytics
h. Cloud SLA (Cloud-Service Level Agreement)

UNIT-I

Q2 Categorically differentiate between SPI Model in cloud computing and Traditional IT Model. Briefly discuss the characteristic features associated with each of these along with schematic illustrations. (10)

Q3 Discuss the significance of Green Cloud Computing and how it envisages to minimize the overall carbon footprint. (10)

UNIT-II

Q4 Elucidate the significance of hypervisor in virtualization. Differentiate between Type-1 and Type-2 hypervisors in terms of CPU performance and security. (10)

Q5 Conceptually, highlight the significance of Cloud Optimization. Discuss the pivotal role it played in boosting cloud performance and discovering cloud cost saving opportunities. (10)

UNIT-III

Q6 What is the role of Identity Access Management (IAM) in cloud security? How do "adopt a role-per-group model" and "grant least privilege" help in achieving the threshold cloud security? (10)

Q7 Categorically differentiate between Static Load Balancing and Dynamic Load Balancing highlighting the significance of server communication and traffic diversion in cloud computing. (10)

UNIT-IV

Q8 Discuss the significance of Meghraj (Cloud Computing initiative of Govt. of India) in accelerating the delivery of e-services while optimizing the ICT expenditure of the government. State its characteristic features and security attributes. (10)

Q9 Critically examine the impact of Cloud Computing in Educational Institutions. Analyze the case-study of "Moodle" or "BYJUs" in revolutionizing the concept of e-education. (10)